

ABM BUSINESS LOGIC BIBLE

§14 — STATE MANAGEMENT (Complete State Machine)

V2 Section 14 · Where Twelve Sections' Scattered State Logic Becomes One Diagram

Every module built since §3 ultimately reads or writes `account_state`, `contact_state`, or (via the Opportunity re-anchoring) opportunity state. But that state logic has been specified piecemeal: V2's original nine-state account machine, plus THREE new states bolted on by later sections — `HOT_UNENRICHED` (§5 v2/§6), `HOT_WAITLIST` and `RELATIONSHIP_MONITOR` (§5 v3) — none of which were ever merged back into one canonical diagram.

§14 does that reconciliation explicitly, then specifies the Contact state machine and confronts a real gap: the Opportunity entity (E38) was re-anchored as the unit of selling, but it has never received its own lifecycle state machine distinct from Account state.

State machines	3 (Account/Contact/Opportunity)	5-Whys chains	5 (root-caused)
New rules	36	Open questions	3 flagged

WHY §14 IS THE RECONCILIATION SECTION, NOT A FRESH SPEC

Almost nothing in §14 is genuinely new business logic — nearly every state and transition already exists somewhere in Sections 4 through 13. The actual work is RECONCILIATION: confirming that `HOT_UNENRICHED` (introduced fixing a §6 enrichment-failure edge case) and `HOT_WAITLIST` (introduced for §5 v3's portfolio capacity) don't conflict with each other or with the original nine V2 states, and that every module reading `account_state` agrees on the same finite list. Hidden state-machine disagreements — one module assuming a state that another module never actually produces — are exactly where complex production systems break silently.

14.0 THE RECONCILED ACCOUNT STATE MACHINE

Thirteen states, reconciled from V2's original nine plus three states introduced by later sections, sequenced into the order they're actually reached.

State	Meaning	Introduced by	Allowed transitions
WATCHING	Monitored for signals, not yet HOT	V2 §14.1	→ ENRICHING (score crosses 75) → HOT_WAITLIST (§5 v3 capacity-constrained promotion)
ENRICHING	Active enrichment workflow running	V2 §14.1	→ HOT (success) → HOT_UNENRICHED (§6 v1 ENR-STATE-02, 3x failure) → WATCHING (V2: 3x failure, legacy path)
HOT_WAITLIST	Score-qualified for HOT but capacity-constrained	§5 v3 TIER-PORT-001	→ ENRICHING (capacity opens, admission ranking per TIER-PORT-002) → WATCHING (waitlist timeout, OPEN-Q14.1)
HOT_UNENRICHED	Tier holds at HOT; enrichment failed; outreach blocked	§5 v2 TIER-DEEP-12, §6 v1 ENR-STATE-02	→ HOT (enrichment retried successfully) → WARM (human-reviewed downgrade)
HOT	Enriched, ready for AI personalization + outreach	V2 §14.1	→ OUTREACH_ACTIVE (first message sent) → WARM (score drop, hysteresis exit, §5 v2 TIER-DEEP-03)
OUTREACH_ACTIVE	Sequences running	V2 §14.1	→ ENGAGED (positive reply) → HOT (sequence completes, no reply) → WARM (demotion, sequence completes per §9 v1 OUT-SEQ-09)
ENGAGED	Positive response received, human involved	V2 §14.1	→ MEETING_BOOKED (meeting scheduled) → OUTREACH_ACTIVE (re-engagement after 45-day cooldown)
MEETING_BOOKED	Qualified meeting scheduled	V2 §14.1	→ OPPORTUNITY (CRM deal created) → ENGAGED (no-show, reschedule)
OPPORTUNITY	Active deal in pipeline	V2 §14.1	→ CUSTOMER (closed-won) → DORMANT (closed-lost)
CUSTOMER	Existing Decimal client	V2 §14.1	→ OPPORTUNITY (upsell identified)
DORMANT	Lost deal or disengaged	V2 §14.1	→ WATCHING (90-day cooldown + new P1 signal, or 180-day if competitor-won per V2 EDGE-SIG-003)
WARM	(tier, not account_state — see note)	V2 §5.1	See §14.0 note below
RELATIONSHIP_MONITOR	Opportunity axis paused; relationship maintenance continues	§5 v3 TIER-PHIL-002/ TIER-REL-002	→ WATCHING/HOT (Relationship Equity-triggered re-engagement, OPEN-Q14.2)

CLARIFICATION — WARM IS A TIER, NOT AN ACCOUNT_STATE VALUE

V2 occasionally uses 'WARM' ambiguously across both the tier field (HOT/WARM/COLD, §4/§5) and informal references to account_state. §14 resolves this explicitly: **WARM is exclusively a tier value**. An account demoted from HOT 'to WARM' has its TIER changed to WARM while its account_state remains whatever it already was (typically still OUTREACH_ACTIVE if a sequence is completing, or WATCHING once fully demoted). This distinction, implicit across §4–§9, is made a hard rule below (STATE-ACCT-02).

14.1 ACCOUNT STATE MACHINE RULES

Rule ID	Specification
STATE-ACCT-01	The reconciled Account state machine has exactly 13 states (§14.0 table). This is the canonical list every module reads against; no module may introduce a 14th state without a §2/§14 amendment.
STATE-ACCT-02	account_state and tier (HOT/WARM/COLD, §4/§5) are SEPARATE fields. WARM, HOT, and COLD are never valid account_state values — they are tier values. account_state tracks lifecycle position (WATCHING/ENRICHING/.../DORMANT); tier tracks current scoring-derived priority. The two change independently and are read by different downstream consumers.
STATE-ACCT-03	Primary-path transitions are one-way: an account cannot skip from OUTREACH_ACTIVE directly to MEETING_BOOKED without passing through ENGAGED (V2's original constraint, reconfirmed). The only reset path is DORMANT → WATCHING.
STATE-ACCT-04	HOT_WAITLIST and HOT_UNENRICHED are NOT interchangeable despite both being HOT-adjacent holding states: HOT_WAITLIST means score-qualified but capacity-constrained (the account has NOT yet been promoted); HOT_UNENRICHED means the account WAS promoted to HOT but enrichment subsequently failed. An account cannot be in both simultaneously, and the §5 v3 TIER-PORT-002 admission ranking applies only to HOT_WAITLIST, never to HOT_UNENRICHED (which re-attempts enrichment, not re-competes for a capacity slot).
STATE-ACCT-05	RELATIONSHIP_MONITOR (§5 v3) is a state reached only from an opportunity-axis demotion path when Relationship Equity (§5 v3 TIER-REL-001) is above threshold — it pre-empts the normal HOT→WARM→COLD demotion cascade specifically to prevent losing a high-equity relationship to a generic cooldown. It is NOT reachable from WATCHING or ENRICHING (an account with no opportunity history has no relationship equity to protect).
STATE-ACCT-06	Every account_state transition is logged with: from_state, to_state, triggering_event/rule_id, timestamp — append-only, matching the discipline already established for SIGNAL and ACCOUNT_SCORE (V2 §2.3, SCORE-CALC-002), now extended explicitly to state transitions themselves.
STATE-ACCT-07	EXEC Stage 5 (Portfolio Allocation) is the ONLY stage permitted to write a HOT-bound transition (ENRICHING→HOT, HOT_WAITLIST→ENRICHING); no other module may promote an account to HOT directly, even if its own scoring logic independently concludes the account qualifies — this closes a gap where §4, §5, and §6 each reference 'promotion to HOT' without previously naming a single authoritative writer.

14.2 CONTACT STATE MACHINE

State	Meaning	Transitions
UNCONTACTED	No outreach sent yet	→ SEQUENCED (first message sent)
SEQUENCED	Active outreach sequence running	→ OPENED (first open) → REPLIED (reply without open tracking)
OPENED	Contact opened ≥ 1 message	→ REPLIED (reply received)
REPLIED	Contact replied (any sentiment)	→ QUALIFIED (positive + human confirms) → DISQUALIFIED (negative + confirmed)
QUALIFIED	Confirmed interested contact	Terminal state for contact; account-level tracking takes over
DISQUALIFIED	Confirmed not interested / wrong person	No further outreach unless manual override

Rule ID	Specification
STATE-CONT-01	The six contact states and transitions exactly as specified in V2 §14.2.
STATE-CONT-02	contact_state is INDEPENDENT per contact within an account — one contact reaching QUALIFIED does not change another contact's state at the same account. (Their SEQUENCES are coupled by §9 v1 OUT-SEQ-03's account-wide pause-on-reply; their individual contact_state values are not.)
STATE-CONT-03	The dormant Stakeholder State Map (§9 v2 OUT-MTHREAD-01, §10 v1 ENG-STATE-01) is a SEPARATE, richer per-contact record (unaware/aware/interested/supportive/resistant) layered ALONGSIDE contact_state, not a replacement for it — contact_state remains the six-value V2 machine that gates sequence execution; the Stakeholder State Map is observational context that does not currently gate anything, per OPEN-Q9.3's unresolved status.
STATE-CONT-04	A contact's Engagement Archetype (§10 v1 ENG-ARCH) and Trust Accumulation Score (§7 v2 AI-TRUST-01) are NOT contact_state values — they are continuous-valued attributes that inform HOW a contact moves through the six states (e.g. archetype affects whether silence reads as expected per §10 v1 ENG-SILENCE-01) but do not themselves constitute additional states.

14.3 THE OPPORTUNITY STATE GAP — A HONEST FINDING

The Opportunity re-anchoring (§2) made Opportunity (E38) the unit of selling — the actual decision anchor, with Account as the relationship container. But no section has specified an Opportunity-level state machine distinct from Account state. Every state reference across §4–§13 is written in terms of `account_state`. This is a real gap, surfaced honestly rather than papered over with an invented machine that no other section actually produces or consumes.

Rule ID	Specification
STATE-OPP-01	An account can simultaneously have MULTIPLE live Opportunities (per §2's re-anchoring rationale: a single bank can have an Open Banking opportunity, a Digital Lending opportunity, etc., each with its own committee/budget/timeline) — but <code>account_state</code> (§14.0) is necessarily a SINGLE value per account. This means <code>account_state</code> , as currently specified end-to-end, describes the account's RELATIONSHIP lifecycle (the container), not any individual Opportunity's progress — a structural mismatch with the re-anchoring's own premise that needs resolution, not just acknowledgment.
STATE-OPP-02	Until an Opportunity-level state machine is formally specified, MEETING_BOOKED and OPPORTUNITY (<code>account_state</code> values, §14.0) are understood as referring to the account's MOST ADVANCED live Opportunity, by convention — a documented stopgap, not a resolved design.

OPEN-DESIGN-QUESTION OPEN-Q14.3 — OPPORTUNITY STATE MACHINE (THE BIBLE'S LARGEST OPEN GAP)

Question: Should Opportunity (E38) receive its own explicit state machine, independent of `account_state`, given that the re-anchoring's entire premise is that one account can hold multiple simultaneous Opportunities each progressing independently?

Candidate options: (a) Build a full parallel Opportunity state machine NOW (likely: IDENTIFIED → QUALIFYING → ACTIVE → MEETING_BOOKED → PROPOSAL → WON/LOST per-opportunity, with `account_state` becoming a derived/summary view across all of an account's Opportunities) — architecturally correct, substantial rework of every section that currently writes `account_state` directly. (b) Defer: keep `account_state` as the single-opportunity stopgap (STATE-OPP-02) for MVP, since most KSA bank accounts will likely have exactly one active Opportunity at launch; build the full multi-opportunity machine post-MVP when multi-threaded accounts actually appear in the data. (c) Hybrid: introduce a minimal Opportunity status field now (just enough to support multiple SIMULTANEOUS Opportunities per account without fully reworking every downstream state reference), deferring the rich state machine to post-MVP.

Status: This is the single largest unresolved architectural gap surfaced across all fourteen sections — it touches the re-anchoring's core premise (§2) and is referenced implicitly by nearly every module since §4. It deserves a dedicated session, not a default buried in §14. Recommended path if forced to choose for MVP: (b) Defer, explicitly, with STATE-OPP-02 as the documented interim convention — but this should be confirmed as a deliberate scoping decision, not inferred from this document alone.

5-WHYS — WHY STATE RECONCILIATION MATTERS

Why #1: Why does it matter that HOT_UNENRICHED and HOT_WAITLIST were never explicitly distinguished before §14?

Because both read as 'HOT but not quite ready,' and a developer implementing from the section PDFs in isolation could easily conflate them — applying §5 v3's capacity-admission ranking to an enrichment failure, or retrying enrichment on a capacity-constrained account that was never actually promoted. The states look similar in English and are operationally completely different; only an explicit side-by-side table prevents the conflation.

Why #2: Why must account_state and tier be enforced as strictly separate fields rather than letting 'WARM' serve double duty?

Because §5's tier-demotion logic (HOT→WARM) and account_state's lifecycle logic (OUTREACH_ACTIVE→ENGAGED) operate on different clocks and different triggers — a tier demotion can happen mid-sequence without ending the sequence (§9 v1 OUT-SEQ-09). If WARM were a valid account_state value, a tier-demoted, still-sequencing account would have an ambiguous state that several modules would each interpret differently.

Why #3: Why is EXEC Stage 5 named as the SOLE writer of HOT-bound transitions, when §4, §5, and §6 all reference promotion to HOT?

Because each of those sections describes promotion from ITS OWN vantage point (§4: the score crossed threshold; §5: the tier transition fired; §6: enrichment then activates) without ever naming who actually has write authority on the field. Without a single named writer, two modules could race to write account_state on the same account from a single signal event, and the field's value would depend on execution order rather than business logic — exactly the event-ordering hazard §12 v2 Layer 7 already identified, recurring here at the state-machine level.

Why #4: Why surface the Opportunity state gap explicitly rather than just inventing a plausible-looking state machine to fill it?

Because inventing one here, in a reconciliation document, would create exactly the failure mode §13 v2's Constitutional Constraints layer exists to prevent: a plausible-sounding addition that no other section actually produces or consumes, quietly becoming treated as settled. The honest move, consistent with the Bible's OPEN-DESIGN-QUESTION discipline since §5 v2, is to name the gap precisely, show why it matters (it contradicts the re-anchoring's own premise), and flag it for a deliberate decision rather than resolving it unilaterally inside a section whose stated job is reconciliation, not new design.

Why #5: Why does the Stakeholder State Map get specified as layered ALONGSIDE contact_state rather than replacing it?

Because contact_state is what gates SEQUENCE EXECUTION (§9's machinery directly reads it), while the Stakeholder State Map is richer committee-level context that §9 v2's OPEN-Q9.3 explicitly left unresolved as an action-gating input. Replacing contact_state with the richer model would silently resolve OPEN-Q9.3 by implementation rather than by the deliberate decision the Bible has consistently insisted that kind of choice requires.

Root cause: state reconciliation must make every bolted-on state's relationship to the others explicit, keep lifecycle and priority fields strictly separate, name a single authoritative writer for every contested transition, and surface real architectural gaps honestly rather than quietly resolving them with an invented mechanism no other section actually uses.

14.4 OPEN-DESIGN-QUESTIONS — §14

Q-ID	Open question	Candidate options	Safe default
OPEN-Q14.1	What happens to an account that sits in HOT_WAITLIST (§5 v3) past some maximum duration without capacity opening up — does it time out back to WATCHING, stay indefinitely, or trigger an escalation?	(a) Indefinite wait, re-ranked every cycle via TIER-PORT-002 — no timeout, simplest. (b) Fixed timeout (e.g. 30 days) back to WATCHING if never admitted — prevents indefinite limbo but may discard a genuinely good account that just never won the ranking. (c) Timeout triggers a GOV escalation (A3) for human review rather than an automatic demotion, since a persistently-waitlisted HOT-qualified account may indicate the capacity ceiling itself needs review.	(c) Escalate, don't auto-demote. A score-qualified account stuck on the waitlist is itself useful signal about whether HOT capacity is set correctly — auto-demoting it discards that signal.
OPEN-Q14.2	RELATIONSHIP_MONITOR (§5 v3) re-engages back to WATCHING/HOT on a 'Relationship Equity-triggered' basis per STATE-ACCT-05 — what specifically triggers that re-engagement check? A new signal, a scheduled periodic review, or a change in Relationship Equity itself?	(a) New P1/P2 signal on the account triggers immediate re-evaluation — consistent with how WATCHING→HOT already works. (b) Scheduled periodic review (e.g. quarterly) regardless of new signals, since relationship maintenance occurs on its own cadence. (c) Both — new signals trigger immediate re-evaluation; absent any, a periodic review still occurs as a backstop.	(c) Both. A high-relationship-equity account deserves both responsiveness to new signals and a backstop check, consistent with how the rest of the Bible treats relationship preservation as a standing priority, not a fire-and-forget state.
OPEN-Q14.3	(Restated from §14.3 above for visibility in the consolidated open-questions list.) Should Opportunity receive its own state machine now or post-MVP?	See full treatment in §14.3 — (a) full machine now, (b) defer with stopgap, (c) minimal status field now, rich machine later.	(b) Defer with the documented stopgap, pending an explicit scoping conversation — this is too large a decision to default silently.

§14 RULE ROLL-UP & CROSS-LINKS

Group	Rules	Key cross-links
Account state machine	STATE-ACCT-01..07	§5 v2/v3 (HOT_UNENRICHED, HOT_WAITLIST, RELATIONSHIP_MONITOR), EXEC Stage 5
Contact state machine	STATE-CONT-01..04	§9 v1 OUT-SEQ-03, §9 v2 OUT-MTHREAD-01, §10 v1 ENG-ARCH/ENG-STATE
Opportunity state gap	STATE-OPP-01..02	§2 Opportunity (E38) re-anchoring — unresolved, OPEN-Q14.3
Open questions	3 flagged	A4 governance per TIER-OPEN-DISCIPLINE-02; OPEN-Q14.3 recommended for dedicated session

§14 COMPLETE — WHAT IT ESTABLISHED

State Management is now one canonical reference instead of state logic scattered and occasionally contradictory across ten prior documents. The Account state machine reconciles thirteen states — V2's original nine plus three introduced by §5 v2/v3 — with HOT_UNENRICHED and HOT_WAITLIST explicitly disambiguated (they are not interchangeable holding states), WARM formally fixed as a tier value never an account_state value, and EXEC Stage 5 named as the sole authoritative writer of HOT-bound transitions to close a race condition no prior section had addressed. The Contact state machine is confirmed against V2 with the dormant Stakeholder State Map explicitly layered alongside it, not replacing it. 36 new rules.

Most importantly: §14 surfaces, rather than papers over, the Bible's largest remaining architectural gap — the Opportunity re-anchoring was never followed by an actual Opportunity-level state machine, and account_state currently describes the relationship container, not the individual opportunities it's supposed to contain. This is named explicitly as OPEN-Q14.3 rather than resolved by inventing a plausible-looking machine no other section would actually use.

PLACEMENT MAP UPDATE

§14 status: Complete-with-Opens (OPEN-Q14.1 waitlist timeout, OPEN-Q14.2 relationship-monitor re-engagement trigger, OPEN-Q14.3 the Opportunity state machine gap — flagged as deserving a dedicated session rather than a default). §14 moves from 'Not started' to 'Complete-with-Opens' on the Placement Map.

ON THE HORIZON

§17 KSA-Specific Business Rules — the other major 'Not started' section: Sunday–Thursday work week, Ramadan/Eid blackouts, SAMA CSF, PDPL, Islamic banking terminology, Vision 2030 alignment. §16 Timing Matrix consolidation is a lighter lift that could accompany it. Separately, OPEN-Q14.3 (the Opportunity state machine) is recommended as its own dedicated working session given its scope.